Transformer-based Symbolic Music Generation

Independent Undergraduate Research Project
Python / PyTorch / MIDI / MidiTok / music21

422 MIDI arrangements | ~40M parameters | 4 representation variants

I independently built a symbolic music generation pipeline to investigate how musical token representations affect a Transformer trained on a limited Touhou corpus. The work covered data preparation, model implementation, training, MIDI generation and empirical evaluation.

GitHub: Archmagos-Lama/text-to-midi-transformer

RESEARCH QUESTIONS

Can a model trained on a small, domain-specific corpus reproduce reference music statistics? How do explicit bar-position and harmonic-role information change the generated distributions?

WHAT I BUILT

• **Data pipeline:** training/validation preparation, pitch-transposition augmentation and REMI tokenization.

• **Model and generation:** an approximately 40M-parameter decoder-only Transformer trained from scratch in PyTorch, with unconditional generation and prompt-based continuation.

• **Evaluation:** 50 generated samples per variant, compared with reference MIDI files using eight musical statistics.

REPRESENTATION EXPERIMENTS

Baseline: retained the standard REMI Bar_None marker. **BARPOS:** replaced it with cyclic position-indexed tokens, BARPOS_0 to BARPOS_127. Equal vocabulary sizes supported a controlled comparison of bar-position encoding.

HarmRole v1: inserted root, third, fifth, seventh or non labels after pitch tokens using music21 bar-level chord estimates. Original files were analysed once; chord maps were transposed for augmented files.

HarmRole v2: split the residual non category into scale and chromatic labels using estimated key membership. This refinement tested a more detailed annotation scheme.

Results & Reflection

Statistical similarity, annotation analysis and practical limitations

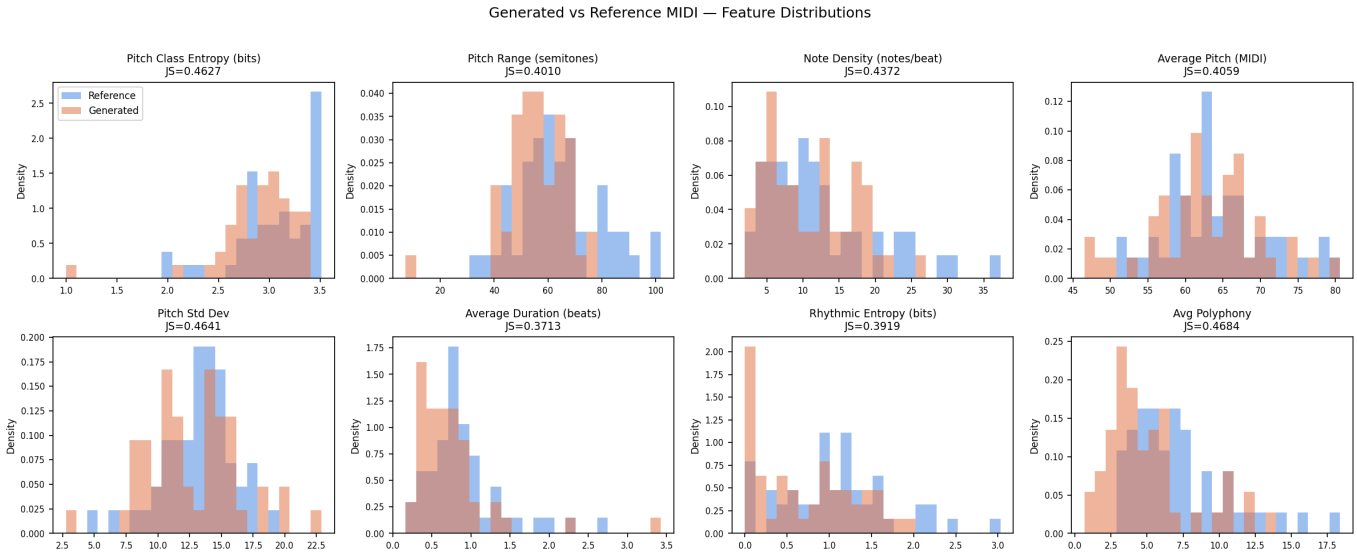
WHAT THE EXPERIMENTS SHOWED

BARPOS: some generated feature distributions were closer to the reference than the baseline, while others were not. The results did not establish an improvement in long-range musical structure.

Annotation finding: 73.8% of v1 harmonic-role labels were non. In v2 this category became 36.4% scale and 37.4% chromatic. This exposed a need to examine annotation assumptions, including chord estimation, bar alignment and global key detection.

Mixed outcome: v2 achieved a lower raw validation loss than v1 (1.4375 versus 1.5050), but generated musical statistics did not improve consistently. Different label vocabularies make raw loss an imperfect cross-variant comparison.

EXPERIMENTAL EVIDENCE



BARPOS generated versus reference MIDI. Comparison across eight musical feature distributions. Blue represents reference samples; orange represents generated samples. Overlap and remaining differences describe statistical similarity, without establishing perceptual quality or long-range musical coherence.

LIMITS OF THE EVIDENCE

Pitch-class entropy, pitch range, note density, average pitch, pitch standard deviation, average duration, rhythmic entropy and polyphony describe MIDI statistics. They do not directly measure motif development, phrase organisation, tension-resolution or perceptual coherence.

Informal sample inspection revealed repetitive rhythms and unstable prompt continuation. This was not a formal listening study. The limited corpus, lack of repeated training runs, and vocabulary and sequence-length differences in HarmRole also restrict causal conclusions.

WHAT I LEARNED

I learned to turn a musical hypothesis into an implemented experiment, inspect unexpected labels, revise a representation and report mixed results. The project suggests that explicit conditioning needs meaningful annotation and an evaluation aligned with the intended musical benefit.